

A latent clinical-anatomical dimension relating metabolic syndrome to brain structure and cognition

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Abstract

The link between metabolic syndrome (MetS) and neurodegenerative as well cerebrovascular conditions holds substantial implications for brain health in at-risk populations. This study elucidates the complex relationship between MetS and brain health by conducting a comprehensive examination of cardiometabolic risk factors, cortical morphology, and cognitive function in 40,087 individuals. Multivariate, data-driven statistics identified a latent dimension linking more severe MetS to widespread brain morphological abnormalities, accounting for up to 71% of shared variance in the data. This dimension was replicable across sub-samples. In a mediation analysis we could demonstrate that MetS-related brain morphological abnormalities mediated the link between MetS severity and cognitive performance in multiple domains. Employing imaging transcriptomics and connectomics, our results also suggest that MetS-related morphological abnormalities are linked to the regional cellular composition and macroscopic brain network organization. By leveraging extensive, multi-domain data combined with a dimensional stratification approach, our analysis provides profound insights into the association of MetS and brain health. These findings can inform effective therapeutic and risk mitigation strategies aimed at maintaining brain integrity.

eLife assessment

This **important** work contributes to our understanding of the combined effects of metabolic syndrome on fronto-temporal gray matter morphology from two large-scale datasets. The evidence based on state-of-the art multivariate imaging analysis and detailed micro- and macrostructural contextualization analyses is **convincing** and provides an understanding of the neurological correlates of metabolic syndrome, although the study would have benefitted from the inclusion of longitudinal data.

1 Introduction

Metabolic syndrome (MetS) represents a cluster of cardiometabolic risk factors, including abdominal obesity, arterial hypertension, dyslipidemia, and insulin resistance [1]. With a prevalence of 23-35% in Western societies, it poses a considerable health challenge, promoting neurodegenerative and cerebrovascular diseases such as cognitive decline, dementia, and stroke [2–6]. As lifestyle and pharmacological interventions can modify the trajectory of MetS, advancing our understanding of its pathophysiological effects on brain structure and function as potential mediators of MetS-related neurological diseases is crucial to inform and motivate risk reduction strategies [7].

Magnetic resonance imaging (MRI) is a powerful non-invasive tool for examining the intricacies of neurological conditions in vivo. Among studies exploring MetS and brain structure, one of the most consistent findings has been alterations in cortical grey matter morphology [8]. Still, our understanding of the relationship between MetS and brain structure is constrained by several factors. To date, there have been only few studies on MetS effects on grey matter integrity that are well-powered [9–12]. The majority of analyses are based on small sample sizes and report effects only on global measures of brain morphology or a priori-defined regions of interest, limiting their scope [12–14]. As a result, reported effects are heterogeneous and most likely difficult to reproduce [15]. Existing large-scale analyses on the isolated effects of individual risk factors (such as hypertension or obesity) do not account for the high covariance of MetS components driven by interacting pathophysiological effects, which may prevent them from capturing the whole picture of MetS as a risk factor composite [16–19]. In addition, analyses addressing the complex interrelationship of MetS, brain structure and cognitive functioning by investigating them in conjunction are scarce [8]. Lastly, while previous studies adopted a case-control design treating MetS as a broad diagnostic category [10–12], a dimensional approach viewing MetS as a continuum could offer a more nuanced representation of the multivariate, continuous nature of the risk factor composite.

Despite reports on MetS effects on brain structure, the determinants and spatial effect patterns remain unclear. A growing body of evidence shows that spatial patterns of brain pathology are shaped by multi-scale neurobiological processes, ranging from the cellular level to regional dynamics to large-scale brain networks [20]. Accordingly, disease effects can not only be driven by local properties, when local patterns of tissue composition predispose individual regions to pathology, but also by topological properties of structural and functional brain networks [20, 21]. Guided by these concepts, multi-modal and multi-scale analysis approaches could advance our understanding of the mechanisms influencing MetS effects on cortical morphology.

We argue that further research leveraging extensive clinical and brain imaging data is required to explore MetS effects on brain morphology. These examinations should integrate 1) a research methodology that strikes a balance between resolving the multivariate connection of MetS and

brain structure while accounting for the high covariance of MetS components; 2) the recognition of impaired cognitive function as a pertinent consequence of MetS; and 3) the analysis of the spatial effect pattern of MetS and its possible determinants.

To meet these research needs, we investigated cortical thickness and subcortical volumetric measurements in a pooled sample of two large-scale population-based cohorts from the UK Biobank (UKB) and Hamburg City Health Study (HCHS) comprising in total 40,087 participants. Partial least squares correlation (PLS) analysis was employed to characterize MetS effects on regional brain morphology. PLS is especially suitable for this research task as it identifies overarching latent relationships by establishing a data-driven multivariate mapping between MetS components and brain morphometric indices. Furthermore, capitalizing on the cognitive phenotyping of both investigated cohorts, we examined the interrelation between MetS, cognitive function and brain structure in a mediation analysis. Finally, to uncover factors associated to brain region-specific MetS effects, we mapped local cellular as well as network topological attributes to observed MetS-associated cortical abnormalities. With this work, we aimed to advance the understanding of the fundamental principles underlying the neurobiology of MetS.

2 Materials and methods

2.1 Study population – the UK Biobank and Hamburg City Health Study

Here, we investigated cross-sectional clinical and imaging data from two large-scale population-based cohort studies: 1) the UK Biobank (UKB, $n = 39,668$, age 45–80 years; application number 41655) and 2) the Hamburg City Health Study (HCHS, $n = 2637$, age 45–74 years) [22, 23]. Both studies recruit large study samples with neuroimaging data alongside a detailed demographic and clinical assessment. Respectively, data for the first visit including a neuroimaging assessment were included. Individuals were excluded if they had a history or a current diagnosis of neurological or psychiatric disease. Field IDs of the used UKB variables are presented in *supplementary table S1*. UKB individuals were excluded based on the non-cancer illnesses codes (<http://biobank.ndph.ox.ac.uk/showcase/coding.cgi?id=6>). Excluded conditions were Alzheimer's disease; alcohol, opioid and other dependencies; amyotrophic lateral sclerosis; brain injury; brain abscess; chronic neurological problem; encephalitis; epilepsy; haemorrhage; head injury; meningitis; multiple sclerosis; Parkinson's disease; skull fracture. Same criteria were applied on HCHS individuals based on the neuroradiological evaluation and self-reported diagnoses variables. To enhance comparability to previous studies we supplemented a case-control analysis enabling to complement continuous multivariate statistical analyses by group statistics. Therefore, a MetS sample was identified based on the consensus definition of the International Diabetes Federation (*supplementary text S2*) and matched to a control cohort.

2.2 Ethics approval

The UKB was ethically approved by the North West Multi-Centre Research Ethics Committee (MREC). Details on the UKB Ethics and Governance framework are provided online (<https://www.ukbiobank.ac.uk/media/0xsbfmw/egf.pdf>). The HCHS was approved by the local ethics committee of the Landesärztekammer Hamburg (State of Hamburg Chamber of Medical Practitioners, PV5131). Good Clinical Practice (GCP), Good Epidemiological Practice (GEP) and the Declaration of Helsinki were the ethical guidelines that governed the conduct of the HCHS [24]. Written informed consent was obtained from all participants investigated in this work.

2.3 Clinical assessment

In the UK Biobank, a battery of cognitive tests is administered, most of which represent shortened and computerized versions of established tests aiming for comprehensive and concise assessment of cognition [25]. From this battery we investigated tests for executive function and processing speed (Reaction Time Test, Symbol Digit Substitution Test, Tower Rearranging Test, Trail Making Tests A and B), memory (Numeric Memory Test, Paired Associate Learning Test, Prospective Memory Test) and reasoning (Fluid Intelligence Test, Matrix Pattern Completion Test). Detailed descriptions of the individual tests can be found elsewhere [26]. Furthermore, some tests (Matrix Pattern Completion Test, Numeric Memory Test, Paired Associate Learning Test, Symbol Digit Substitution Test, Trail Making Test and Tower Rearranging Test) are only administered to a subsample of the UKB imaging cohort explaining the missing test results for a subgroup of participants.

In the HCHS, cognitive testing was administered by a trained study nurse and included the Animal Naming Test, Trail Making Test A and B, Verbal Fluency and Word List Recall subtests of the Consortium to Establish a Registry for Alzheimer's Disease Neuropsychological Assessment Battery (CERAD-Plus), as well as the Clock Drawing Test [27,28].

2.4 MRI acquisition

The full UKB neuroimaging protocol can be found online (https://biobank.ctsu.ox.ac.uk/crystal/crystal/docs/brain_mri.pdf) [22]. MR images were acquired on a 3-T Siemens Skyra MRI scanner (Siemens, Erlangen, Germany). T1-weighted MRI used a 3D MPRAGE sequence with 1-mm isotropic resolution with the following sequence parameters: repetition time = 2000 ms, echo time = 2.01 ms, 256 axial slices, slice thickness = 1 mm, and in-plane resolution = 1 x 1 mm. In the HCHS, MR images were acquired as well on a 3-T Siemens Skyra MRI scanner. Measurements were performed with a protocol as described in previous work [24]. In detail, for 3D T1-weighted anatomical images, rapid acquisition gradient-echo sequence (MPRAGE) was used with the following sequence parameters: repetition time = 2500 ms, echo time = 2.12 ms, 256 axial slices, slice thickness = 0.94 mm, and in-plane resolution = 0.83 x 0.83 mm.

2.5 Estimation brain morphological measures

To achieve comparability and reproducibility, the preconfigured and containerized CAT12 pipeline (CAT12.7 r1743; <https://github.com/m-wierzbka/cat-container>) was employed for surface reconstruction and cortical thickness measurement building upon a projection-based thickness estimation method as well as computation of subcortical volumes [29]. Cortical thickness measures were normalized from individual to 32k fsLR surface space (conte69) to ensure vertex correspondence across subjects. Subcortical volumes were computed for the Melbourne Subcortex Atlas parcellation resolution 1 [30]. Volumetric measures for the anterior and posterior thalamus parcel were averaged to obtain a single measure for the thalamus. Individuals with a CAT12 image quality rating lower than 75% were excluded during quality assessment. To facilitate large-scale data management while ensuring provenance tracking and reproducibility, we employed the DataLad-based FAIRly big workflow for image data processing [31].

2.6 Statistical analysis

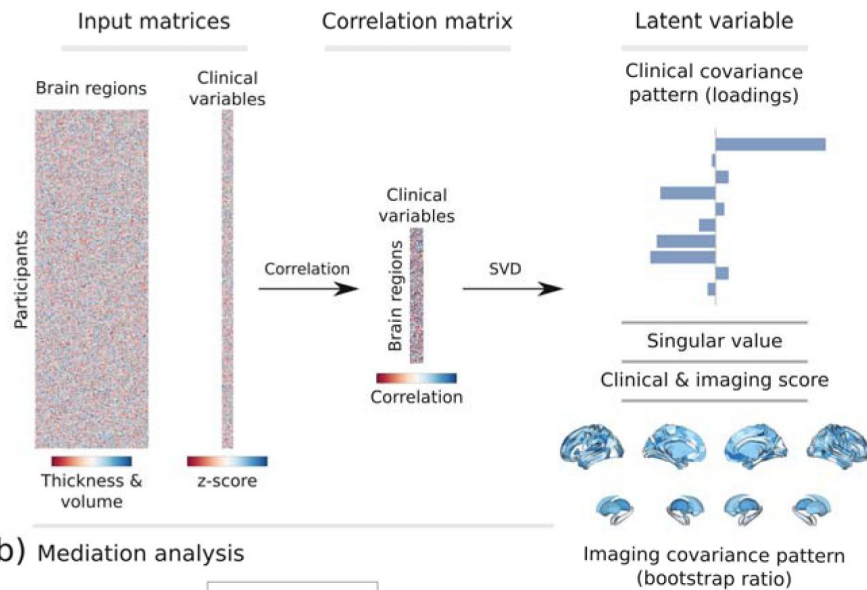
Statistical computations and plotting were performed in python 3.9.7 leveraging bctpy (v. 0.6.0), brainstat (v. 0.3.6), brainSMASH (v. 0.11.0), the ENIGMA toolbox (v. 1.1.3), matplotlib (v. 3.5.1), neuromaps (v. 0.0.1), numpy (v. 1.22.3), pandas (v. 1.4.2), pingouin (v. 0.5.1), pyts (v. 0.0.1), scikit-learn (v. 1.0.2), scipy (v. 1.7.3), seaborn (v. 0.11.2) as well as in matlab (v. 2021b) using ABAnnotate (v. 0.1.1).

2.6.1 Partial least squares correlation analysis

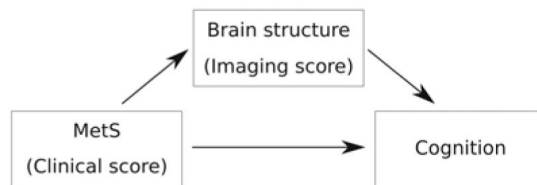
To relate MetS components and cortical morphology, we performed a PLS using `pyls` (<https://github.com/rmarkello/pyls>). PLS identifies covariance profiles that relate two sets of variables in a data-driven double multivariate analysis [32]. Here, we related regional cortical thickness and subcortical volumes to clinical measurements of MetS components, i.e., obesity (waist circumference, hip circumference, waist-hip ratio, body mass index), arterial hypertension (systolic blood pressure, diastolic blood pressure), dyslipidemia (high density lipoprotein, low density lipoprotein, total cholesterol, triglycerides) and insulin resistance (HbA1c, non-fasting blood glucose). Before conducting the PLS, missing values were imputed via k-nearest neighbor imputation ($n_{\text{neighbor}} = 4$) with imputation only taking into account variables of the same group, i.e., MetS component variables were imputed based on the remaining MetS component data only and not based on demographic variables. To account for age, sex, education and cohort (UKB/HCHS) as potential confounds, they were regressed out of brain morphological and MetS component data.

We then performed PLS as described in previous work [33]. Methodological details are covered in **figure 1a** and *supplementary text S3*. Brain morphological measures were randomly permuted ($n_{\text{random}} = 5000$) to assess statistical significance of derived latent variables and their corresponding covariance profiles. Subject-specific PLS scores, including a clinical score and an imaging score, were computed. Higher scores indicate stronger adherence to the respective covariance profiles: a high clinical score signifies pronounced expression of the clinical profile, and a high imaging score reflects marked adherence to the brain morphological profile. Bootstrap resampling ($n_{\text{bootstrap}} = 5000$) was performed to assess the contribution of individual variables to the imaging-clinical relationship. Confidence intervals (95%) of singular vector weights were computed for clinical variables to assess the significance of their contribution. To estimate contributions of brain regions, bootstrap ratios were computed as the singular vector weight divided by the bootstrap-estimated standard error. A high bootstrap ratio is indicative of a region's contribution, as a relevant region shows a high singular vector weight alongside a small standard error implying stability across bootstraps. The bootstrap ratio equals a z-score in case of a normally distributed bootstrap. Hence, brain region contributions were considered significant if the bootstrap ratio was >1.96 or <-1.96 (95% confidence interval). Overall model robustness was assessed via a 10-fold cross-validation by correlating out-of-sample PLS scores within each fold.

a) Partial least squares correlation



b) Mediation analysis



c) Contextualization analysis

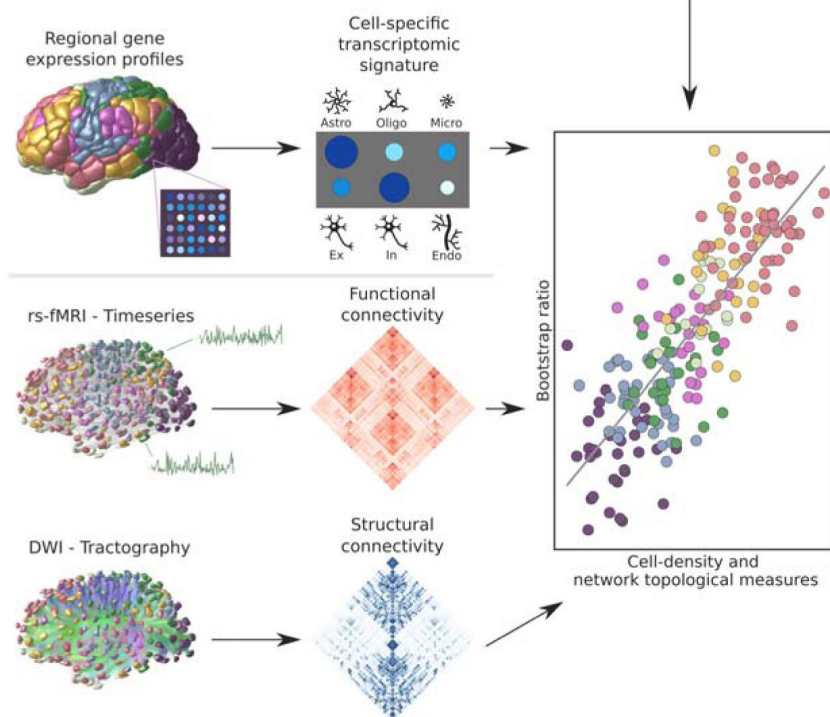


Figure 1.

Methodology.

a) Illustration of the partial least squares correlation analysis. Starting from two input matrices containing per-subject information of regional morphological measures as well as clinical data (demographic and MetS-related risk factors) a correlation matrix is computed. This matrix is subsequently subjected to singular value decomposition resulting in a set of mutually orthogonal latent variables. Latent variables each consist of a left singular vector (here, clinical covariance profile), singular value and right singular vector (here, imaging covariance profile). In addition, subject-specific clinical and imaging scores are computed. b) The interplay between MetS, brain structure and cognition was investigated in a post-hoc mediation analysis. We tested whether the relationship between the clinical score, representing MetS severity, and different cognitive test performances was statistically mediated by the imaging score. c) Contextualization analysis. Upper row: based on microarray gene expression data, the densities of different cell populations across the cortex were quantified. Middle and lower row: based on functional and structural group-consensus connectomes based on data from the Human Connectome Project, metrics of functional and structural brain network topology were derived. Cell density as well as connectomic measures were related to the bootstrap ratio via spatial correlations. Modified from Petersen et al. and Zeighami et al. [33,82].

Abbreviations: Astro – astrocytes; DWI – diffusion-weighted magnetic resonance imaging; Endo – endothelial cells; Ex – excitatory neuron populations (Ex1-8); In – inhibitory neuron populations (In1-8); Micro – microglia; Oligo – oligodendrocytes; rs-fMRI – resting-state functional magnetic resonance imaging; SVD – singular value decomposition.

2.6.2 Mediation analysis

In a post-hoc mediation analysis, we investigated how the subject-specific clinical PLS score of the first latent variable, reflecting the degree of an individual's expression of the identified MetS risk profile, relates to cognitive test outcomes, and whether this relationship is influenced by the imaging PLS score of the first latent variable, which represents the degree of brain morphological differences (**figure 1b**). This analysis allows to separate the total effect of the clinical PLS score on cognitive performance into: (1) a direct effect (the immediate impact of clinical scores on cognition), and (2) an indirect effect (the portion influenced by the imaging PLS score). This approach helps to disentangle the complex interplay between MetS and cognitive function by examining the role of brain structural effects as a potential intermediary. We considered an indirect effect as mediating if there was a significant association between the clinical and imaging PLS scores, the imaging PLS score was significantly associated to the cognitive outcome, and if the link between clinical scores and cognitive outcomes weakened (partial mediation) or became insignificant (full mediation) after accounting for imaging scores. The significance of mediation was assessed using bootstrapping ($n_{\text{bootstrap}}=5000$), with models adjusted for age, sex, and education. To obtain standardized estimates, mediation analysis inputs were z-scored beforehand. Given the variation in cognitive test batteries between the UKB and HCHS cohorts, only individuals with results from the respective tests were considered in each mediation analysis. To account for the different versions of the Trail Making Tests A and B used in both cohorts, test results were harmonized through z-scoring within the individual subsamples before pooled z-scoring.

2.6.3 Contextualization analysis

We investigated the link of MetS and regional brain morphological measurements in the context of cell-specific gene expression profiles and structural and functional brain network characteristics (**figure 1c**). Therefore, we used the Schaefer-parcellated (400×7 and 100×7, v.1) bootstrap ratio

map and related it to indices representing different gene expression and network topological properties of the human cortex via spatial correlations (Spearman correlation, r_{sp}) on a group-level [34].

Virtual histology analysis. We performed a virtual histology analysis leveraging gene transcription information to quantify the density of different cell populations across the cortex employing the ABAnnotate toolbox [35,36]. Genes corresponding with specific cell populations of the central nervous system were identified based on a classification derived from single nucleus-RNA sequencing data [37]. The gene-celltype mapping is provided by the PsychENCODE database (http://resource.psychencode.org/Datasets/Derived/SC_Decom/DER-19_Single_cell_markersgenes_TPM.x/sx) [38]. The abagen toolbox (v. 0.1.3) was used to obtain regional microarray expression data of these genes for Schaefer100×7 parcels based on the Allen Human Brain Atlas (AHBA) [39]. The Schaefer100×7 atlas was used as it better matches the sampling density of the AHBA eventually resulting in no parcels with missing values. Regional expression patterns of genes corresponding to astrocytes, endothelial cells, excitatory neuron populations (Ex1-8), inhibitory neuron populations (In1-8), microglia, and oligodendrocytes were extracted. Instead of assessing the correspondence between MetS effects and the expression pattern of each gene directly, we employed ensemble-based gene category enrichment analysis (GCEA) [40]. This approach represents a modification to customary GCEA addressing the issues of gene-gene dependency through within-category co-expression which is caused by shared spatial embedding as well as spatial autocorrelation of cortical transcriptomics data. In brief, gene transcription indices were averaged within categories (here cell populations) and spatially correlated with the bootstrap ratio map. Statistical significance was assessed by comparing the empirical correlation coefficients against a null distribution derived from surrogate maps with preserved spatial embedding and autocorrelation computed via a spatial lag model [41]. Further details on the processing steps covered by ABAnnotate can be found elsewhere (<https://osf.io/gcxun>) [42].

Brain network topology. To investigate the cortical MetS effects pattern in the context of brain network topology, three connectivity metrics were leveraged based on data from structural and functional brain imaging: weighted degree centrality, neighborhood abnormality as well as macroscale functional connectivity gradients as described previously [33]. These were computed based on functional and structural consensus connectomes on group-level derived from the Human Connectome Project Young Adults dataset comprised in the ENIGMA toolbox [43,44]. Computation and derivation of the metrics are described in the *supplementary text S4*. For this analysis, statistical significance of spatial correlations was assessed via spin permutations ($n = 1,000$) which represent a null model preserving the inherent spatial autocorrelation of cortical information [45]. Spin permutations are performed by projecting parcel-wise data onto a sphere which then is randomly rotated. After rotation, information is projected back on the surface and a permuted r_{sp} is computed. A p-value is computed comparing the empirical correlation coefficient to the permuted distribution. To assure that our results do not depend on null model choice, we additionally tested our results against a variogram-based null model implemented in the brainSMASH toolbox (<https://github.com/murraylab/brainsmash>) as well as a network rewiring null model with preserved density and degree sequence [46,47].

All p-values resulting from both contextualization analyses were FDR-corrected for multiple comparisons. As we conducted this study mindful of the reuse of our resources, the MetS effect maps are provided as separate supplementary files to enable further analyses.

2.6.4 Sensitivity analyses

For a sensitivity analysis, we reperformed the PLS separately within the UKB and HCHS cohorts. In contrast to the PLS main analysis, in these subset specific PLS analyses cognitive test performances were also incorporated as clinical variables as cognitive batteries were subset specific. This

approach was employed to evaluate the stability of the results and to determine if cognitive tests contribute to the latent variables.

To test whether the PLS indeed captures the link of MetS and brain morphology, we conducted a group comparison as in previous studies of MetS. Besides descriptive group statistics, the cortical thickness of individuals with MetS and matched controls was compared on a surface vertex-level leveraging the BrainStat toolbox (v 0.3.6, <https://brainstat.readthedocs.io/>) [48]. A general linear model was applied correcting for age, sex, education and cohort effects. Vertex-wise p-values were FDR-corrected for multiple comparisons. To demonstrate the correspondence between the t-statistic and cortical bootstrap ratio maps, we related them via spatial correlation analyses. The t-statistic map was also used for sensitivity analysis of the virtual histology analysis and brain network contextualization.

To ensure that the brain network contextualization results were not biased by the connectome choice, we reperformed the analysis with structural and functional group consensus connectomes based on resting-state functional and diffusion-weighted MRI data from the HCHS. The corresponding connectome reconstruction approaches were described elsewhere [33].

3 Results

3.1 Sample characteristics

Application of exclusion criteria and quality assessment ruled out 2,188 UKB subjects and 30 HCHS subjects resulting in a final analysis sample of 40,087 individuals. For a flowchart providing details on the sample selection procedure please refer to *supplementary figure S5*. Descriptive statistics are listed in *table 1*. To sensitivity analyze our results, as well as to facilitate the comparison with previous reports which primarily rely on a case-control design, we supplemented group statistics comparing individuals with clinically defined MetS and matched controls, where applicable. Corresponding group analysis results are described in more detail in *supplementary materials S6-12*.

3.2 Partial least squares correlation analysis

We investigated the relationship between brain morphological and clinical measures of MetS (abdominal obesity, arterial hypertension, dyslipidemia, insulin resistance) in a PLS considering all individuals from both studies (n=40,087). By this, we aimed to detect the continuous effect of any MetS component independent from a formal binary classification of MetS (present / not present). A correlation matrix relating all considered MetS component measures is displayed in *supplementary figure S13*. Before conducting the PLS, brain morphological and clinical data were deconfounded for age, sex, education and cohort effects.

PLS identified seven significant latent variables which represent clinical-anatomical dimensions relating MetS components to brain morphology (*supplementary table S14*). The first latent variable explained 71.20% of shared variance and was thus further investigated (*figure 2a*). Specifically, the first latent variable corresponded with a covariance profile of lower severity of MetS (*figure 2c*; loadings [95% confidence interval]; waist circumference: -.230 [-.239,-.221], hip circumference: -.187 [-.195,-.178], waist-hip ratio: -.167 [-.176,-.158], body mass index: -.234 [-.243,-.226], systolic blood pressure: -.089 [-.098,-.080], diastolic blood pressure: -.116 [-.125,-.107], high density lipoprotein: .099 [.090,.108], low density lipoprotein: -.013 [-.022,-.004], total cholesterol: .003 [-.006,.012], triglycerides: -.102 [-.111,-.092], HbA1c: -.064 [-.073,-.054], glucose: -.049 [-.058,-.039]). Notably, the obesity-related measures showed the strongest contribution to the covariance profile as indicated by the highest loading to the latent variable. Age (<.001 [-.009,.009]), sex (<.001 [-.009,.009]), education (<.001 [-.009,.009]) and cohort (<.001 [-.008,.007]) did not

Metric	Stat ^a
Age (years)	63.55 ± 7.59 (40087)
Sex (% female)	46.47 (40087)
Education (ISCED)	2.62 ± 0.73 (39944)
<u>Metabolic syndrome components</u>	
Waist circumference (cm)	88.47 ± 12.71 (38800)
Hip circumference (cm)	100.90 ± 8.79 (38801)
Waist-hip ratio	0.88 ± 0.09 (38800)
Body mass index	26.47 ± 4.37 (38701)
RR _{systolic} (mmHg)	138.30 ± 18.57 (31234)
RR _{diastolic} (mmHg)	78.88 ± 10.09 (31238)
Antihypertensive therapy (%)	6.96 (39976)
HDL (mg/dL)	61.76 ± 23.69 (34468)
LDL (mg/dL)	137.38 ± 36.29 (37456)
Cholesterol (mg/dL)	211.29 ± 56.42 (37531)
Triglycerides (mg/dL)	148.90 ± 83.84 (37510)
Lipid lowering therapy (%)	14.44 (39976)
HbA1c (%)	5.37 ± 0.48 (37284)
Blood glucose (mg/dL)	90.29 ± 17.58 (34432)
Antidiabetic therapy (%)	0.45 (39976)

Table 1.

Descriptive statistics UKB and HCHS

<u>Imaging</u>	
Mean cortical thickness (mm)	2.40 ± 0.09 (40087)
<u>Cognitive variables of the UK Biobank</u>	
Fluid Intelligence	6.63 ± 2.06 (36510)
Matrix Pattern Completion	7.99 ± 2.13 (25771)
Numeric Memory Test	6.69 ± 1.52 (26780)
Paired Associate Learning	6.92 ± 2.63 (26048)
Prospective Memory	1.07 ± 0.39 (37192)
Reaction Time (sec)	594.16 ± 109.08 (37015)
Symbol Digit Substitution	18.96 ± 5.25 (25810)
Tower Rearranging Test	9.91 ± 3.23 (25555)
Trail Making Test A (sec)	223.03 ± 86.51 (26048)
Trail Making Test B (sec)	550.01 ± 270.09 (26048)
<u>Cognitive variables of the Hamburg City Health Study</u>	
Animal Naming Test	24.78 ± 6.92 (2416)
Clock Drawing Test	6.43 ± 1.12 (2479)
Trail Making Test A (sec)	40.09 ± 14.33 (2290)
Trail Making Test B (sec)	90.05 ± 37.30 (2264)
Multiple-Choice Vocabulary Intelligence Test	31.27 ± 3.58 (2026)
Word List Recall	7.75 ± 1.84 (2342)
Abbreviations: cm = centimeter, dL = deciliter, HDL = high density lipoprotein, ISCED = International	

Table 1. (continued)

Standard Classification of Education, mg = milligram, mm = millimeters, mmHg = millimeters of mercury, RR = Blood pressure, sec = seconds
^a Presented as mean ± SD (N)

Table 1. (continued)

significantly contribute to the latent variable, which is compatible with sufficient effects of deconfounding. Details on the second latent variable which explained 22.33% of shared variance are provided in *supplementary figure S15*. In brief, it predominantly related lower HbA1c and blood glucose to higher thickness and volume in lateral frontal, posterior temporal, parietal and occipital regions and vice versa.

Bootstrap ratios ($= \frac{\text{singular vector weight}}{\text{bootstrap-estimated standard error}}$) were computed to identify brain regions with a significant contribution to the covariance profile (see Methods). Cortical thickness in orbitofrontal, lateral prefrontal, insular, anterior cingulate and temporal areas as well as all volumes of all investigated subcortical regions contributed positively to the covariance profile as indicated by a positive bootstrap ratio (*figure 2d*). Thus, a higher cortical thickness and subcortical volume in these areas corresponded with less obesity, hypertension, dyslipidemia and insulin resistance and vice versa, i.e., lower cortical thickness and subcortical volumes with increased severity of MetS. A negative bootstrap ratio was found in superior frontal, parietal and occipital regions indicating that a higher cortical thickness in these regions corresponded with a more pronounced expression of MetS components. This overall pattern was confirmed via conventional, vertex-wise group comparisons of cortical thickness measurements based on the binary classification of individuals with MetS and matched controls (*supplementary figure S12*) as well as subsample analyses considering the UKB and HCHS participants independently (*supplementary figure S16-17*). The correlation matrix of all spatial effect maps investigated in this study (bootstrap ratio and Schaefer400-parcellated t-statistic from group comparisons) is visualized in *supplementary figure S18*. All derived effect size maps were significantly correlated ($r_{sp} = .67 - .99$, $p_{FDR} < .05$) [*34*].

Subject-specific imaging and clinical scores for the first latent variable were computed. These scores indicate to which degree an individual expresses the corresponding covariance profiles. By definition, the scores are correlated ($r_{sp} = .201$, $p < .005$, *figure 2b*) indicating that individuals exhibiting the clinical covariance profile (severity of MetS components) also express the brain morphological pattern. This relationship was robust across a 10-fold cross-validation (avg. $r_{sp} = .19$, *supplementary table S19*).

These results were consistent in separate PLS analyses for both the UKB and HCHS samples, as displayed in *supplementary figures S16 and S17*. In these subset-specific analyses, cognitive test performances significantly contributed to the first latent variable when included in the PLS. Consequently, the first latent variable associated more severe MetS with both brain morphological abnormalities and poorer cognitive performance.

3.3 Mediation analysis of cognitive outcomes

To gain a better understanding of the link between MetS, brain morphology, and cognitive function, we performed a mediation analysis on cognitive test results and subject-specific PLS scores. Therefore, we investigated whether the imaging PLS score (representing MetS-related brain structural abnormalities) acts as a mediator in the relationship between the clinical PLS score (representing MetS severity) and cognitive test performances. Importantly, scores of the main PLS analysis, which did not include cognitive measures, were considered. The corresponding path plots are shown in *figure 3*. The imaging score was found to fully mediate the relationship of the clinical score and results of the Trail Making Test B ($ab = -.011$, $p_{FDR} < .001$; $c' = -.012$, $p_{FDR} = .072$; $c = .023$, $p_{FDR} < .001$), Fluid Intelligence Test ($ab = .017$, $p_{FDR} < .001$; $c' = .011$, $p_{FDR} = .072$; $c = .028$, $p_{FDR} < .001$) as well as Matrix Pattern Completion Test ($ab = .015$, $p_{FDR} < .001$; $c' = .010$, $p_{FDR} = .172$; $c = .025$, $p_{FDR} < .001$). Further, the imaging score partially mediated the relationship of the clinical score and results of the Symbol Digit Substitution Test ($ab = .010$, $p_{FDR} < .001$; $c' = .036$, $p_{FDR} < .001$; $c = 0.046$, $p_{FDR} < .001$), Numeric Memory Test ($ab = .014$, $p_{FDR} < .001$; $c' = .044$, $p_{FDR} < .001$; $c = 0.058$, $p_{FDR} < .001$) and Paired Associate Learning Test ($ab = .015$, $p_{FDR} < .001$; $c' = .044$, $p_{FDR} < .001$; $c = 0.059$, $p_{FDR} < .001$). For the remaining cognitive tests, no significant mediation was found.

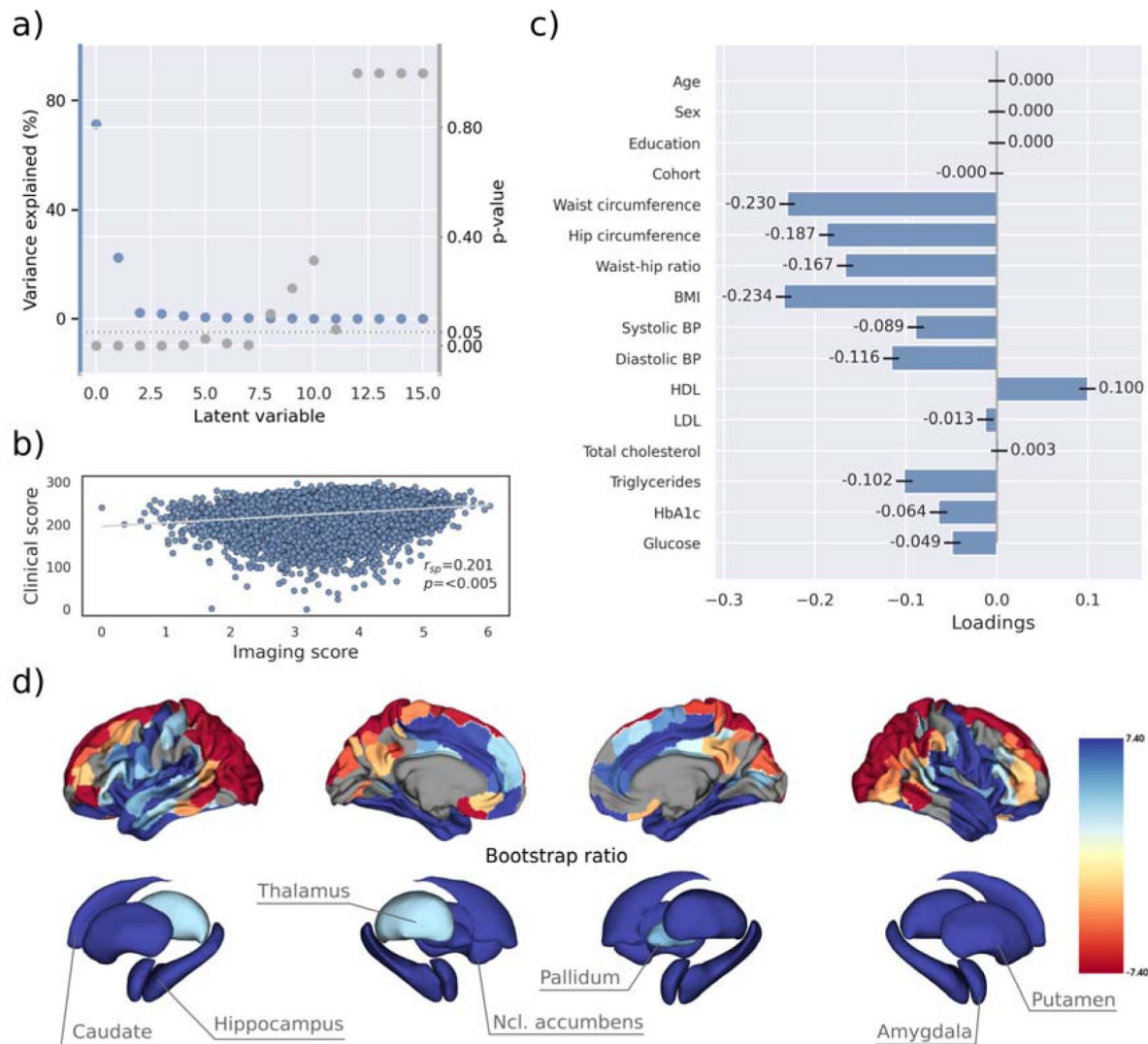


Figure 2.

Partial least squares (PLS) analysis.

a) Explained variance and p-values of latent variables. b) Scatter plot relating subject-specific clinical and imaging PLS scores. Higher scores indicate higher adherence to the respective covariance profile. c) Clinical covariance profile. 95% confidence intervals were calculated via bootstrap resampling. Note that confound removal for age, sex, education and cohort was performed prior to the PLS. d) Imaging covariance profile represented by bootstrap ratio. A high positive or negative bootstrap ratio indicates high contribution of a brain region to the overall covariance profile. Vertices with a significant bootstrap ratio (> 1.96 or < -1.96) are highlighted by colors. Abbreviations: - Spearman correlation coefficient.

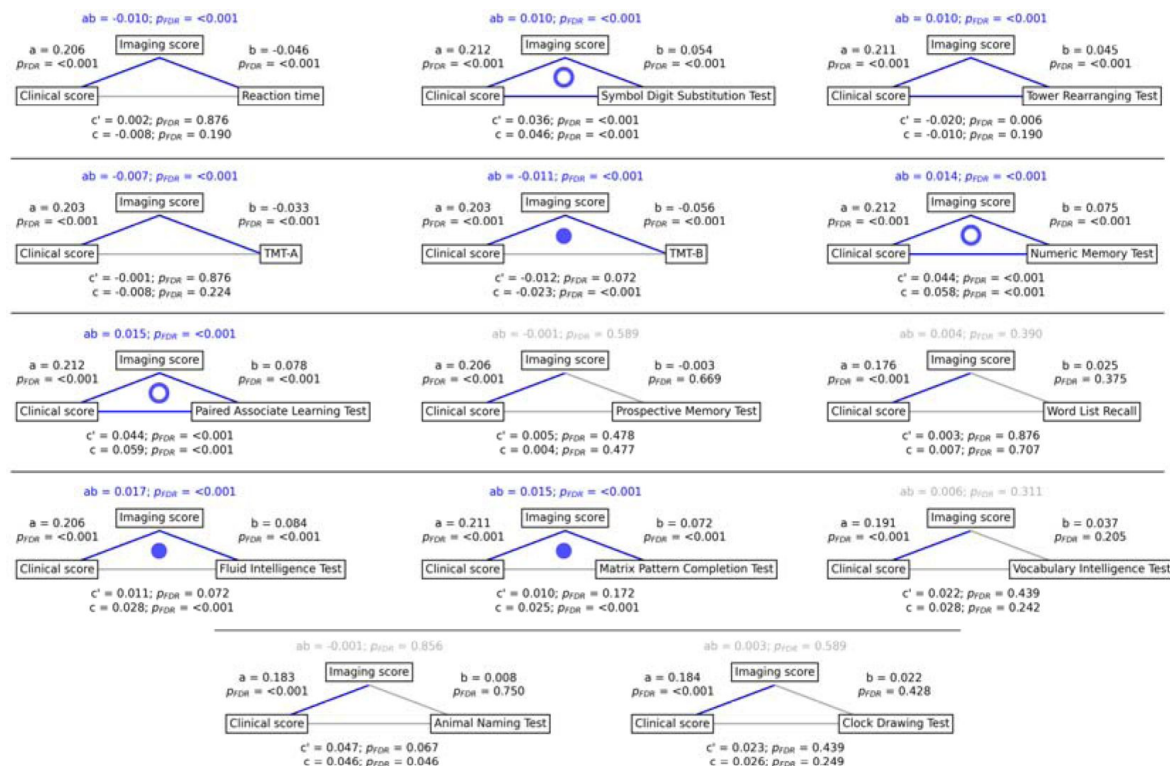


Figure 3.

Mediation analysis results.

Mediation effects of subject-specific imaging PLS scores on the relationship between MetS represented by the clinical PLS score and cognitive test performances. Path plots display standardized effects and p-values: (a) clinical score to imaging score, (b) imaging score to cognitive score, (ab) indirect effect (c') direct effect and (c) total effect. Significant paths are highlighted in blue; non-significant in light gray. If the indirect effect ab was significant, the text for ab is highlighted in blue. A blue dot in the path plot indicates if a relationship is significantly mediated, i.e., the indirect effect ab was significant and the direct effect c' was reduced or non-significant compared to the total effect c. An empty dot indicates a partial mediation, a full dot indicates a full mediation. *Abbreviations:* - false discovery rate-corrected p-values; PLS – partial least squares correlation; TMT-A – Trail Making Test A; TMT-B – Trail Making Test B.

3.4 Contextualization of MetS-associated brain morphological abnormalities

We investigated whether the pattern of MetS effects on cortical structure is conditioned by the regional density of specific cell populations and global brain network topology in a surface-based contextualization analysis (see Methods).

Therefore, we first used a virtual histology approach to relate the bootstrap ratio from PLS to the differential expression of cell-type specific genes based on microarray data from the Allen Human Brain Atlas [49]. The results are illustrated in [figure 4](#). The bootstrap ratio was significantly positively correlated with the density of endothelial cells ($Z_{\text{r-sp}} = .190$, $p_{\text{FDR}} = .016$), microglia ($Z_{\text{r-sp}} = .271$, $p_{\text{FDR}} = .016$), excitatory neurons type 8 ($Z_{\text{r-sp}} = .165$, $p_{\text{FDR}} = .016$), inhibitory neurons type 1 ($Z_{\text{r-sp}} = .363$, $p_{\text{FDR}} = .036$) and excitatory neurons type 6 ($Z = .146$, $p_{\text{FDR}} = .034$) indicating that MetS-related brain morphological abnormalities are strongest in regions of the highest density of these cell types. No significant associations were found with regard to the remaining excitatory neuron types (Ex1-Ex5, Ex7), inhibitory neurons (In2-In8), astrocytes and oligodendrocytes (*supplementary table S20*). Virtual histology analysis results for bootstrap ratios corresponding with latent variables 2 and 3 are shown in *supplementary figure S21*. As a sensitivity analysis, we contextualized the t-statistic map derived from group statistics. The results remained stable except for excitatory neurons type 6 ($Z_{\text{r-sp}} = .145$, $p_{\text{FDR}} = .123$) and inhibitory neurons type 1 ($Z_{\text{r-sp}} = .432$, $p_{\text{FDR}} = .108$), which no longer showed a significant association (*supplementary materials S22-23*).

Second, we associated the bootstrap ratio with three pre-selected measures of brain network topology derived from group consensus functional and structural connectomes of the Human Connectome Project (HCP) ([figure 5](#)): weighted degree centrality (marking brain network hubs), neighborhood abnormality and macroscale functional connectivity gradients [33]. The bootstrap ratio showed a medium positive correlation with the functional neighborhood abnormality ($r_{\text{sp}} = .464$, $p_{\text{spin}} < .001$, $p_{\text{smash}} < .001$, $p_{\text{rewire}} < .001$) and a strong positive correlation with the structural neighborhood abnormality ($r_{\text{sp}} = .764$, $p_{\text{spin}} < .001$, $p_{\text{smash}} < .001$, $p_{\text{rewire}} < .001$) indicating functional and structural interconnectedness of areas exhibiting similar MetS effects. These results remained significant when the t-statistic map was contextualized instead of the bootstrap ratio as well as when neighborhood abnormality measures were derived from consensus connectomes of the HCHS instead of the HCP (*supplementary figure S24-25*). We found no significant associations for the remaining indices of network topology, i.e., functional degree centrality ($r_{\text{sp}} = .163$, $p_{\text{spin}} = .365$, $p_{\text{smash}} = .406$, $p_{\text{rewire}} = .870$), structural degree centrality ($r_{\text{sp}} = .029$, $p_{\text{spin}} = .423$, $p_{\text{smash}} = .814$, $p_{\text{rewire}} = .103$) as well as functional cortical gradient 1 ($r_{\text{sp}} = .152$, $p_{\text{spin}} = .313$, $p_{\text{smash}} = .406$, $p_{\text{rewire}} = .030$) and gradient 2 ($r_{\text{sp}} = -.177$, $p_{\text{spin}} = .313$, $p_{\text{smash}} = .406$, $p_{\text{rewire}} < .001$).

4 Discussion

We investigated the impact of MetS on brain morphology and cognitive function in a large sample of individuals from two population-based neuroimaging studies. We report three main findings: 1) multivariate, data-driven statistics revealed a latent variable relating MetS and brain health: participants were distributed along a clinical-anatomical dimension of interindividual variability, linking more severe MetS to widespread brain morphological abnormalities. Negative MetS-related brain morphological abnormalities were strongest in orbitofrontal, lateral prefrontal, insular, cingulate and temporal cortices as well as subcortical areas. Positive MetS-related brain morphological abnormalities were strongest in superior frontal, parietal and occipital regions. 2) The severity of MetS was associated with executive function and processing speed, memory, and reasoning test performances, and was found to be statistically mediated by MetS-related brain morphological abnormalities. 3) The pattern of MetS-related brain morphological abnormalities

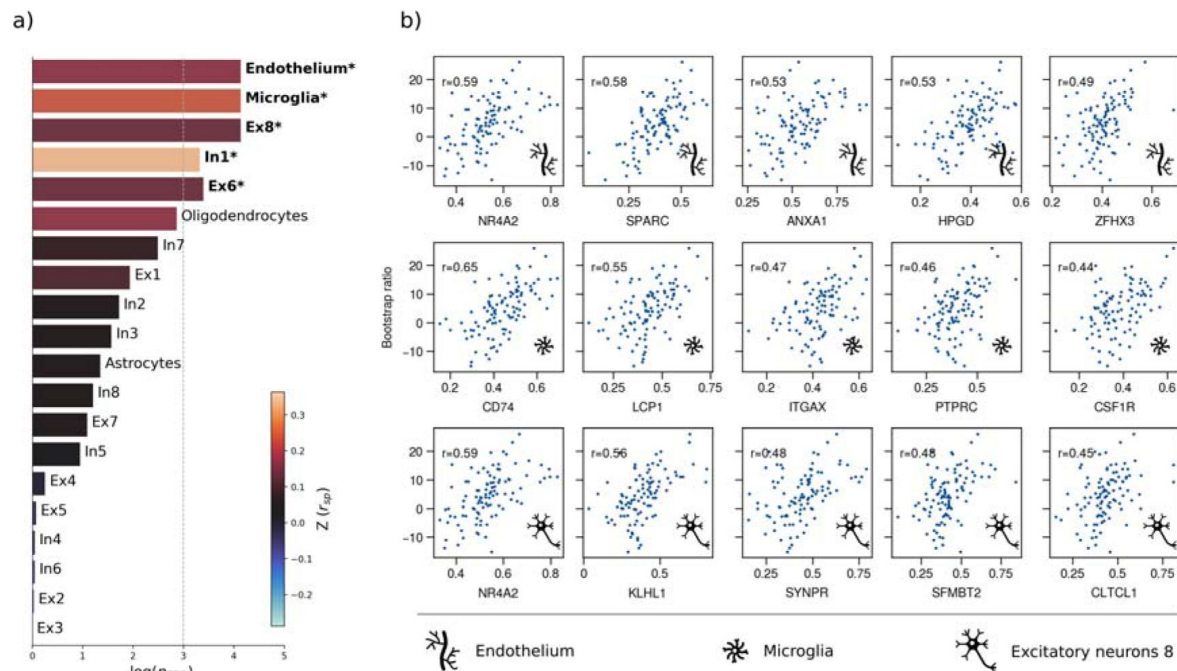


Figure 4.

Virtual histology analysis.

The correspondence between MetS effects (bootstrap ratio) and cell type-specific gene expression profiles was examined via an ensemble-based gene category enrichment analysis. a) Barplot displaying spatial correlation results. The bar height displays the significance level. Colors encode the aggregate z-transformed Spearman correlation coefficient relating the Schaefer100-parcellated bootstrap ratio and respective cell population densities. Asterisks indicate statistical significance. The significance threshold of <0.05 is highlighted by a vertical dashed line. b) Scatter plots illustrating spatial correlations between MetS effects and exemplary cortical gene expression profiles per cell population significantly associated across analyses – i.e., endothelium, microglia and excitatory neurons type 8. Top 5 genes most strongly correlating with the bootstrap ratio map were visualized for each of these cell populations. Icons in the bottom right of each scatter plot indicate the corresponding cell type. A legend explaining the icons is provided at the bottom. First row: endothelium; second row: microglia; third row: excitatory neurons type 8. Virtual histology analysis results for the bootstrap ratios of latent variables 2 and 3 are shown in *supplementary figure S21*. A corresponding plot illustrating the contextualization of the t-statistic derived from group statistics is shown in *supplementary figure S22*. Abbreviations: $-\log(p_{FDR})$ – negative logarithm of the false discovery rate-corrected p-value derived from spatial lag models [36, 41]; r – Spearman correlation coefficient. $Z(r_{sp})$ – aggregate z-transformed Spearman correlation coefficient.

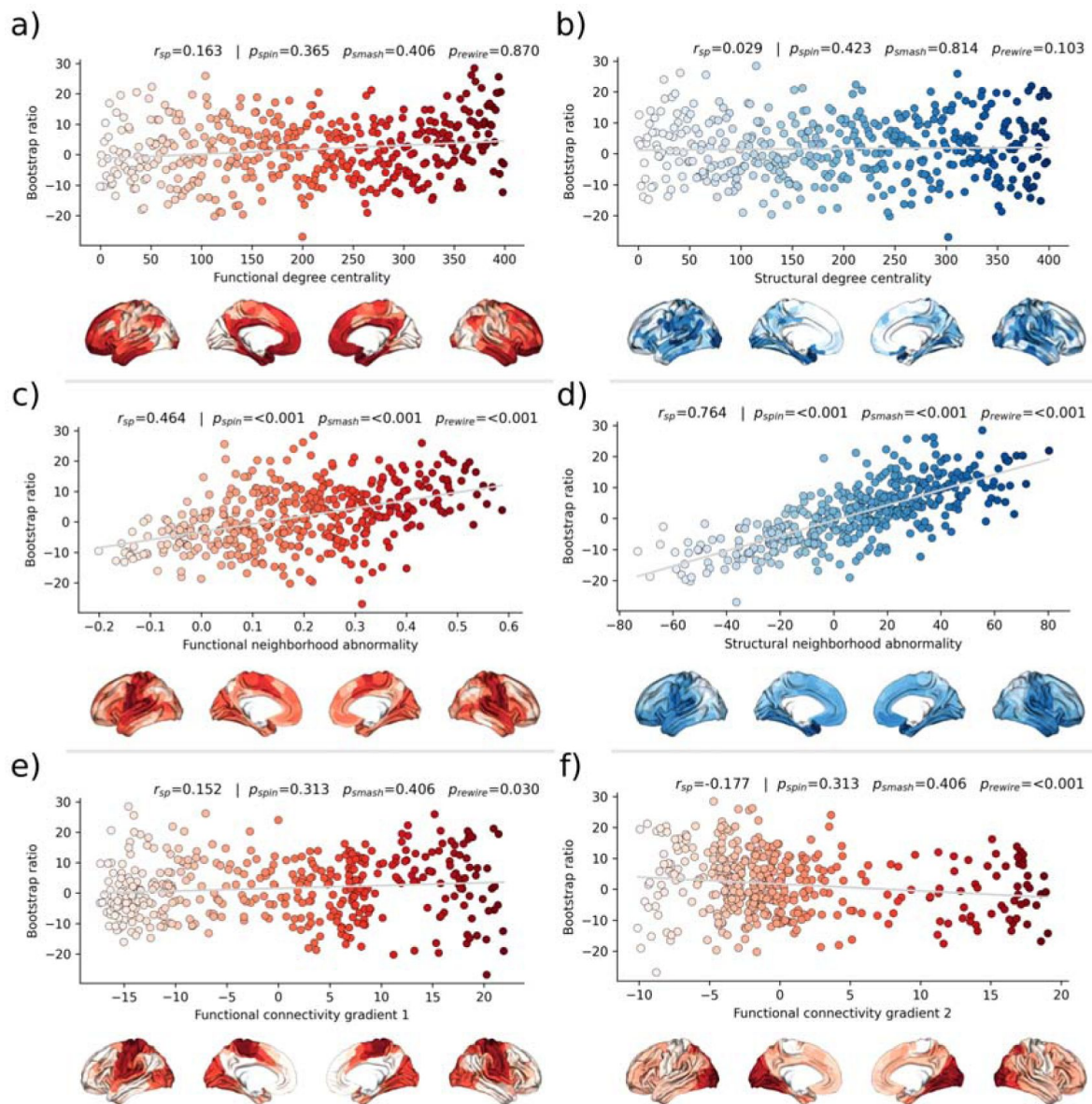


Figure 5.

Brain network contextualization.

Spatial correlation results derived from relating Schaefer400×7-parcellated maps of MetS effects (bootstrap ratio) to network topological indices (red: functional connectivity, blue: structural connectivity). Scatter plots that illustrate the spatial relationship are supplemented by respective surface plots for anatomical localization. The color coding of cortical regions and associated dots corresponds. a) & b) Functional and structural degree centrality rank. c) & d) Functional and structural neighborhood abnormality. e) & f) Intrinsic functional network hierarchy represented by functional connectivity gradients 1 and 2. Complementary results concerning t-statistic maps derived from group comparisons between MetS subjects and controls are presented in *supplementary figure S24*. Corresponding results after reperforming the analysis with HCHS derived group-consensus connectomes are presented in *supplementary figure S25*. Abbreviations: HCHS – Hamburg City Health Study; p_{rewire} - p-value derived from network rewiring [47]; p_{smash} - p-value derived from brainSMASH surrogates [46]; p_{spin} - p-value derived from spin permutation results [45]; r_{sp} - Spearman correlation coefficient.

appeared to be linked to regional cell composition as well as functional and structural connectivity. These findings were robust across sensitivity analyses. In sum, our study provides an in-depth examination of the intricate relationship between MetS, brain morphology and cognition.

4.1 PLS reveals a latent clinical-anatomical dimension relating MetS and brain health

MetS adversely impacts brain health through complex, interacting effects on the cerebral vasculature and parenchyma as shown by histopathological and imaging studies [19]; The pathophysiology of MetS involves atherosclerosis, which affects blood supply and triggers inflammation [50,51]; endothelial dysfunction reducing cerebral vasoreactivity [52]; breakdown of the blood-brain barrier inciting an inflammatory response [53]; oxidative stress causing neuronal and mitochondrial dysfunction [54]; and small vessel injury leading to various pathologies including white matter damage, microinfarcts and cerebral microbleeds [55].

To address these interacting effects, we harnessed multivariate, data-driven statistics in form of a PLS in two large-scale population-based studies to probe for covariance profiles relating the full range of MetS components (such as obesity or arterial hypertension) to regional brain morphological information in a single analysis. PLS identified seven significant latent variables with the first variable explaining the majority (71.20%) of shared variance within the imaging and clinical data (*figure 2a*). This finding indicates a relatively uniform connection between MetS and brain morphology, implying that the associative effects of various MetS components on brain structure are comparatively similar, despite the distinct pathomechanisms each component entails.

PLS revealed that all MetS components were contributing to this latent signature. However, waist circumference, hip circumference, waist-hip ratio and body mass index consistently contributed higher than the remaining variables across conducted analyses which highlights obesity as the strongest driver of MetS-related brain morphological abnormalities.

We interpret these findings as evidence that MetS-associated conditions jointly contribute to the harmful effects on brain structure rather than affecting it in a strictly individual manner. This notion is supported by previous work in the UKB demonstrating overlapping effects of individual risk factors on cortical morphology [56]. Specifically, the first latent variable related increased severity of obesity, dyslipidemia, arterial hypertension and insulin resistance with lower thickness in orbitofrontal, lateral prefrontal, insular, cingulate and temporal cortices as well as lower volume across subcortical regions (*figure 2c and d*). This profile was consistent in separate PLS analyses of UKB and HCHS participants as well as group comparisons (*supplementary figures S12 and S16-17*). Previous research aligns with our detection of a MetS-associated frontotemporal morphometric abnormality pattern [9,13,57]. As a speculative causative pathway, human and animal studies have related the orbitofrontal, insular and anterior cingulate cortex to food-related reward processing, taste and impulse regulation [58,59]. Conceivably, structural alterations of these brain regions are linked to brain functions and behaviors that exacerbate the risk profile leading to MetS [60,61]. We also noted a positive MetS-cortical thickness association in superior frontal, parietal and occipital lobes, a less intuitive finding that has been previously reported [62,63]. Although speculative, the positive effects might be due to MetS compensating cholesterol disruptions associated with neurodegenerative processes [64].

The second latent variable accounted for 22.33% of shared variance and linked higher markers of insulin resistance and lower dyslipidemia to lower thickness and volume in lateral frontal, posterior temporal, parietal and occipital regions. The distinct covariance profile of this latent variable, compared to the first, likely indicates a separate pathomechanistic connection between

MetS components and brain morphology. Given that HbA1c and blood glucose were the most significant contributors to this variable, insulin resistance might drive the observed clinical-anatomical relationship.

4.2 Brain morphological abnormalities mediate the relationship between MetS and cognitive deficits

Cognitive performance has been consistently linked to cardiometabolic risk factors in health and disease [65]. Yet, the pathomechanistic correlates of this relationship remain to be understood. Our mediation analysis revealed that increased MetS severity correlates with worse performance in executive function and processing speed (Symbol Digit Substitution Test, Trail Making Test B), memory (Numeric Memory Test, Paired Associate Learning Test), and reasoning (Fluid intelligence, Matrix Pattern Completion Test), with brain morphological abnormalities statistically mediating these relationships. Additionally, group comparisons indicated poorer cognitive performance in MetS subjects (*supplementary tables S9-10*) and including cognitive outcomes in the PLS as clinical variables revealed a significant contribution to the first latent variable (*supplementary materials S16b and S17b*). These results suggest that MetS is significantly associated with cognitive deficits across various domains, and brain morphological abnormalities are a crucial pathomechanistic link in this relationship. In support of this, previous studies have shown that brain structure mediates the relationship between MetS and cognitive performance in a pediatric sample and elderly patients with vascular cognitive impairment [66–68]. The detected latent variable might represent a continuous disease spectrum spanning from minor cognitive deficits due to a cardiometabolic risk profile to severe cognitive deficits due to dementia. In support of this hypothesis, the determined brain morphological abnormality pattern is consistent with the atrophy pattern found in vascular mild cognitive impairment, vascular dementia and Alzheimer's dementia [67–69].

Collectively, these findings highlight the role of MetS in cognitive impairment and underscore the potential impact of therapies targeting cardiometabolic risk factors. Although the definitive role of such therapies in preventing cognitive decline is not yet fully established, emerging evidence suggests that these interventions can mitigate the adverse cognitive effects of MetS [70–72]. As our results highlight obesity as a key factor in the observed clinical-anatomical relationship, we think that future studies should further investigate weight-reducing interventions to examine their effects on cognitive outcomes. Advanced neuroimaging techniques promise to refine these therapeutic approaches by enabling to identify MetS patients at risk of cognitive decline that would benefit the most from targeted interventions for cognitive health protection.

4.3 MetS-related brain morphological abnormalities link to cellular tissue composition and network topology

To better understand the emergence of the spatial pattern of MetS-related brain morphological abnormalities, we conducted two contextualization analyses leveraging reference datasets of local gene expression data as well as properties of brain network topology.

Using a virtual histology approach based on regional gene expression data, we investigated MetS effects in relation to cell population densities (*figure 4*). As the main finding, we report that higher MetS-related brain morphological abnormalities coincide with a higher regional density of endothelial cells. This aligns with the known role of endothelial dysfunction in MetS compromising tissues via chronic vascular inflammation, increased thrombosis risk and hypoperfusion due to altered vasoreactivity and vascular remodeling [52]. As endothelial density also indicates the degree of general tissue vascularization, well-vascularized regions are also likely more exposed to cardiometabolic risk factor effects in general [50]. Our results furthermore indicate that microglial density determines a brain region's susceptibility to MetS effects. Microglia are resident macrophages of the central nervous system sustaining neuronal

integrity by maintaining a healthy microenvironment. Animal studies have linked microglial activation mediated by blood-brain barrier leakage and systemic inflammation to cardiometabolic risk [73,74]. Activated microglia can harm the brain structure by releasing reactive oxygen species, proinflammatory cytokines and proteinases [75]. Lastly, we found an association with the density of excitatory neurons of subtype 8. These neurons reside in cortical layer 6 and their axons mainly entertain long-range cortico-cortical and cortico-thalamic connections [37,76]. Consequently, layer 6 neurons might be particularly susceptible to MetS effects due to their exposition to MetS-related white matter disease [24,77]. Taken together, the virtual histology analysis indicates that MetS-related brain morphological abnormalities are associated to local cellular fingerprints. Our findings emphasize the involvement of endothelial cells and microglia in brain structural abnormalities due to cardiometabolic risk, marking them as potential targets for therapies aimed at mitigating MetS effects on brain health.

For the second approach, we contextualized MetS-related brain morphological abnormalities using principal topological properties of functional and structural brain networks. We found that regional MetS effects and those of functionally and structurally connected neighbors were correlated (*figure 5c and 5d*) – i.e., areas with similar MetS effects tended to be disproportionately interconnected. Put differently, MetS effects coincided within functional and structural brain networks. Therefore, our findings can be interpreted as evidence that a region's functional and structural network embedding – i.e., its individual profile of functional interactions as well as white matter fiber tract connections – are associated to its susceptibility to morphological MetS effects. Multiple mechanisms might explain how connectivity might be associated to MetS-related morphological alterations. For example, microvascular pathology might impair white matter fiber tracts leading to joint degeneration in interconnected cortical brain areas: that is, the occurrence of shared MetS effects within functionally and structurally connected neighborhoods is explained by their shared (dis-)connectivity profile [78]. In support of this, previous work using diffusion tensor imaging suggests that MetS-related microstructural white matter alterations preferentially occur in the frontal and temporal lobe, which spatially matches the frontotemporal morphometric differences observed in our work [79]. Furthermore, we speculate on an interplay between local and network-topological susceptibility in MetS: functional and structural connectivity may provide a scaffold for propagating MetS-related perturbation across the network in the sense of a spreading phenomenon – i.e., a region might be influenced by network-driven exposure to regions with higher local susceptibility. Observed degeneration of a region might be aggravated by malfunctional communication to other vulnerable regions including mechanisms of excitotoxicity, diminished excitation and metabolic stress [80]. These findings underscore the relevance of brain network organization in understanding the pathomechanistic link of MetS and brain morphology.

While this work's strengths lie in a large sample size, high-quality MRI and clinical data, robust image processing, and a comprehensive methodology for examining the link of MetS and brain health, it also has limitations. First, the virtual histology analysis relies on post-mortem brain samples, potentially different from in-vivo profiles. In addition, the predominance of UKB subjects may bias the results, and potential reliability issues of the cognitive assessment in the UKB need to be acknowledged [81]. Lastly, the cross-sectional design restricts the ability for demonstrating causative effects. Longitudinal assessment of the surveyed relationships would provide more robust evidence and therefore, future studies should move in this direction.

5 Conclusion

Our analysis revealed associative effects of MetS, structural brain integrity and cognition, complementing existing efforts to motivate and inform strategies for cardiometabolic risk reduction. In conjunction, a characteristic and reproducible structural imaging fingerprint associated with MetS was identified. This pattern of MetS-related brain morphological

abnormalities was linked to local histological as well as global network topological features. Collectively, our results highlight how an integrative, multi-modal and multi-scale analysis approach can lead to a more holistic understanding of the neural underpinnings of MetS and its risk components. As research in this field advances, leveraging neuroimaging may improve personalized cardiometabolic risk mitigation approaches.

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Each author has made a significant contribution to the manuscript and all authors read and approved its final version. We describe individual contributions to the paper using the CRediT contributor role taxonomy. Conceptualization: M.P., F.H., B.C.; Data Curation: M.P., F.H., F.N., C.M., M.S.; Formal analysis: M.P.; Funding acquisition: S.B.E., G.T., B.C.; Investigation: M.P.; Methodology: M.P.; Software: M.P., F.H.; Supervision: G.T., B.C.; Visualization: M.P.; Writing—original draft: M.P.; Writing—review & editing: M.P., F.H., F.N., C.M., M.S., L.R., B.Z., T.Z., S.K., J.G., J.F., R.T., A.O., K.P., S.B.E., G.T., B.C.

6 Declaration of interest

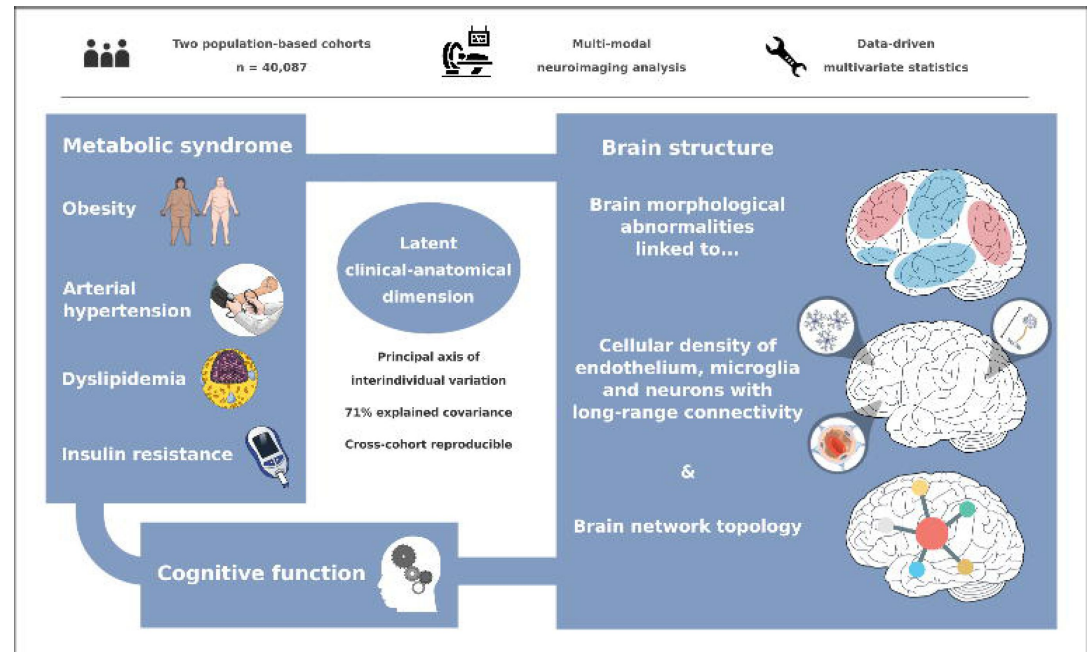
JG has received speaker fees from Lundbeck, Janssen-Cilag, Lilly, Otsuka and Boehringer outside the submitted work. JF reported receiving personal fees from Acandis, Cerenovus, Microvention, Medtronic, Phenox, and Penumbra; receiving grants from Stryker and Route 92; being managing director of eppdata; and owning shares in Tegus and Vastrax; all outside the submitted work. GT has received fees as consultant or lecturer from Acandis, Alexion, Amarin, Bayer, Boehringer Ingelheim, BristolMyersSquibb/Pfizer, Daichi Sankyo, Portola, and Stryker outside the submitted work. TZ and RT are listed as co-inventors of an international patent on the use of a computing device to estimate the probability of myocardial infarction (PCT/EP2021/073193, International Publication Number WO2022043229A1). TZ and RT are shareholders of the company ART-EMIS GmbH Hamburg. The remaining authors declare no conflicts of interest.

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8 Data availability

UK Biobank data can be obtained via its standardized data access procedure (<https://www.ukbiobank.ac.uk/>). HCHS participant data used in this analysis is not publicly available for privacy reasons, but access can be established via request to the HCHS steering committee. The analysis code is publicly available on GitHub (https://github.com/csi-hamburg/2023_petersen_mets_brain_morphology and <https://github.com/csi-hamburg/CSIframe/wiki/Structural-processing-with-CAT>).



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Reviewer #1 (Public Review):

Summary:

In their study, Petersen et al. investigated the relationship between parameters of metabolic syndrome (MetS) and cortical thickness using partial least-squares correlation analysis (PLS) and performed subsequently a group comparison (sensitivity analysis). To do this, they utilized data from two large-scale population-based cohorts: the UK BioBank (UKB) and the Hamburg City Health Study (HCHS). They identified a latent variable that explained 77% of the shared variance, driven by several measures related to MetS, with obesity-related measures having the strongest contribution. Their results highlighted that higher cortical thickness in the orbitofrontal, lateral prefrontal, insular, anterior cingulate, and temporal areas is associated with lower MetS severity. Conversely, the opposite pattern was observed in the superior frontal, parietal, and occipital regions. A similar pattern was then observed in the sensitivity analysis when comparing two groups (MetS vs. matched controls) separately.

Interestingly, after including HbA1c (a blood glycemic marker, which reflects insulin resistance much better than non-fasting glucose) in their revision, the authors identified a second latent variable accounting for 22% of shared variance mostly driven by HbA1c and blood glucose. The authors conclude that the distinct covariance profile of this variable likely indicates a separate pathological mechanistic connection between MetS components and brain morphology.

They then mapped local cellular and network topological attributes to the observed cortical changes associated with MetS. This was achieved using cell-type-specific gene expressions from the Allen Human Brain Atlas and the group consensus functional and structural connectomes of the Human Connectome Project (HCP), respectively. This contextualization analysis allowed them to identify potential cellular contributions in these structures driven by endothelial cells, microglial cells, and excitatory neurons. It also indicated functional and structural interconnectedness of areas experiencing similar MetS effects.

Strengths:

The effects of metabolic syndrome on the brain are still incompletely understood, and such multi-scale analyses are important for the field. Despite the study's sole 'correlation-based' nature, it yields valuable results, including several scales of brain parameters (cortical thickness, cellular, and network-based). The results are robust and benefit from two 'large-scale' datasets, resulting in highly powered statistics

Weaknesses:

The weakness of this study lies mostly on the non-causative approach used here. Nevertheless, the authors are aware of the limitations of the study and carefully frame their language accordingly.

<https://doi.org/10.7554/eLife.93246.2.sa1>

Reviewer #2 (Public Review):

Summary:

In this manuscript, Petersen et al. aimed for a comprehensive assessment of the relationship between cardiometabolic risk factors and cortical thickness. They found that a latent variable reflecting higher obesity, hypertension, LDL cholesterol, triglycerides, non-fasting glucose, HbA1c and lower HDL cholesterol was associated with lower cortical thickness in orbitofrontal, lateral prefrontal, insular, anterior cingulate and temporal areas as well as lower subcortical volumes. In sensitivity analyses they showed that this pattern replicated across cohorts and was also consistent with a clinical definition of the metabolic syndrome.

Further, when including cognition into the multivariate analysis, the pattern remained unchanged and mediation analyses showed that the relationship between the first latent variable and worse cognitive performance across several tests was mediated by the brain morphological differences.

The authors investigated the cell types implicated in the regions associated with cardiometabolic risk using the Allen Brain Atlas and found that the density of excitatory neurons type 8, endothelial cells and microglia reliably co-located with the pattern of cortical thickness. Furthermore, they showed that cortical regions more strongly associated with MetS were more closely structurally & functionally connected than others.

Strengths:

This study performed a comprehensive assessment of the combined association of cardiometabolic risk factors and brain structure and investigated micro-and macroscopic underpinnings. A major strength of the study is the methodological approach of partial least squares which allows one to not single out risk factors but to take them into account simultaneously. The large sample size from two cohorts allowed for different sensitivity analyses and convincing evidence for the stability of the first latent variable. The authors demonstrated that the component was also reliably related to cognitive performance and that the association of the individual cardiometabolic risk on cognition was mediated by brain morphological differences, replicating multiple previous studies which evidenced associations of different components of the MetS with worse cognitive performance.

The novel contribution of the study lies in the virtual histology and brain topology investigation of the cortical pattern related to MetS. The virtual histology provided convincing evidence of the co-localization of endothelial, glial and excitatory neuronal cells with the regions of MetS-associated cortical thinning while the brain topology analysis highlighted the disproportionate structural and functional connectivity between associated regions. This analysis provides insights into the role of inflammatory processes and the intricate link between gray matter morphology and microvasculature, both locally and in relation to long-range connectivity. This information is valuable to inform future mechanistic studies.

Weaknesses:

The study is exclusively cross-sectional which does not allow disentangling potential causes from consequences. While studies indicate that most of the differences seen in middle age are probably consequences of the MetS on the vasculature, blood-brain barrier or inflammatory processes, differences in cortical morphology might also represent a risk factor for weight gain.

The study is exploratory in nature and for the contextualization analyses it is difficult to judge whether those were selected from a larger pool of analyses.

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Author Response

The following is the authors' response to the original reviews.

Reviewer 1

Comment 1.1: "Did the UKB or HCHS datasets have information on accurate markers of insulin resistance, such as HbA1c or HOMA-IR (if fasting glucose was not available)? Looking at that data would allow us to determine the contribution of insulin resistance to the observed cortical phenotype."

Reply 1.1: We appreciate the insightful suggestion from the reviewer. In response, we incorporated the HbA1c into our analysis, enhancing its sensitivity to potential effects of insulin resistance. Subsequently, our analysis was reperformed, integrating HbA1c alongside non-fasting blood glucose in the PLS. This addition did not alter our main results, i.e., that of the PLS, virtual histology, and network contextualization analysis. Notably, as a result of the inclusion of HbA1c, the second latent variable now accounted for a greater shared variance (22.13%), with HbA1c showing the highest loading among MetS component variables. The manuscript has been thoroughly revised to incorporate these results.

Comments 1.2: "(Results, p.13, 291-292) "A correlation matrix relating all considered MetS component measures is displayed in supplementary figure S12. Please clarify in this figure labels whether this was non-fasting glucose. If this is non-fasting glucose, it is not a MetS-related risk factor. The reader might be misled into thinking that fasting-glucose has a weak correlation, while its contribution (and the effect of insulin resistance) was not studied here."

"Table S8 and Table S9: Is the glucose metric here measured following fasting? If not, this should not be listed as a metabolic syndrome criterion. Or it should be specified that it isn't fasted glucose, otherwise, it sounds misleading."

Reply 1.2: We thank the reviewer for bringing this ambiguity to our attention. The initial analysis included only non-fasting plasma glucose in the PLS, as fasting plasma glucose data was unavailable for UKB and HCHS participants. Following your suggestion in reply 1.1, we have now incorporated HbA1c, a more indicative marker of insulin resistance. We retained non-fasting blood glucose in our analysis, recognizing its relevance as a diagnostic variable for type 2 diabetes mellitus, although it is less informative than fasting plasma glucose, HbA1c, or HOMA-IR. This decision is substantiated by the significant correlation found between non-fasting plasma glucose and HbA1c in our sample ($r=.49$).

To enhance clarity, we have revised the methods section to explicitly mention that the study investigates non-fasting blood glucose. The revised sentence reads: "Here, we related regional cortical thickness and subcortical volumes to clinical measurements of MetS components, i.e., obesity (waist circumference, hip circumference, waist-hip ratio, body mass index), arterial

hypertension (systolic blood pressure, diastolic blood pressure), dyslipidemia (high density lipoprotein, low density lipoprotein, total cholesterol, triglycerides) and insulin resistance (HbA1c, non-fasting blood glucose)."

Additionally, we have updated the caption of supplementary figure S13 (formerly supplementary figure S12) to clearly indicate the investigation of non-fasting plasma glucose. The table detailing diagnostic MetS criteria (supplementary table S2) has also been amended to clarify the absence of fasting plasma glucose data in our study and to indicate that only data on antidiabetic therapy and diagnosis of type 2 diabetes mellitus were used as criteria for insulin resistance in the case-control analysis.

Comment 1.3: "I do not understand how the authors can claim there is a deterministic relationship there if all the results are only correlational or comparative. Can the differences in functional connectivity and white matter fiber tracts observed not be caused by the changes in cortices they relate to? How can the authors be sure the network organisation is shaping the cortical effects and not the opposite (the cortical changes influence the network organisation)? This should be further discussed or explained."

Reply 1.3: We agree with the reviewer's comment on the non-causative nature of our data and have accordingly revised the discussion section to reflect a more cautious interpretation of our findings. We have carefully reframed our language to avoid any implications of causality, ensuring the narrative aligns with the correlational nature of our data. Nevertheless, we believe that exploring causal interpretations can offer valuable clinical insights. Therefore, while moderating our language, we have maintained certain speculative discussions regarding potential causative pathomechanistic pathways.

Comment 1.4: "The hippocampus is also an area where changes have consistently been observed. Why did the authors limit their analysis to the cortex."

Reply 1.4: We appreciate this reviewer comment. In response, we have added volumes of Melbourne Subcortical Atlas parcels (including the hippocampus) to the analysis. Corresponding results are now shown in figure 2. The subcortical bootstrap ratios indicated that higher MetS severity was related to lower volumes across all investigated subcortical structures.

Comment 1.5: "Which field ID of the UK biobank are the measures referring to? If possible, please specify the Field ID for each of the UKB metrics used in the study."

Reply 1.5: We thank the reviewer for the recommendation. The Field IDs used in our study are now listed in supplementary figure S1.

Comment 1.6: "Several Figures were wrongly annotated, making it hard to follow the text."

Reply 1.6: Thank you for bringing the annotation issues to our awareness. We have thoroughly edited all annotations which should now correctly reference the figure content.

Reviewer 2

Comment 2.1: "Do the authors have the chance to see how the pattern relates to changes in cognitive function in the UKBB and possibly HCHS? This could help to provide some evidence about the directionality of the effect." Reply 2.1: Thank you for your suggestion. We acknowledge the potential value of investigating gray matter morphometric data alongside longitudinal information on cognitive function. Although we concur with the

significance of this approach, we are constrained by the ongoing processing of the UKB's imaging follow-up data and the pending release of the HCHS follow-up data. Consequently, our current analysis cannot incorporate this aspect for now. We plan to explore the relationship between MetS, cognition and brain morphology using longitudinal data as soon as it becomes available.

Comment 2.2: "Also, you could project new data onto the component and establish a link with cognition in a third sample which would be even more convincing. I can offer LIFE-Adult study for this aim."

Reply 2.2: We are grateful for your recommendation to enhance our study's robustness by including a third sample to establish a cognitive link. While we recognize the merit of such a sensitivity analysis, we believe that our current dataset, derived from two large, independent cohorts, is sufficiently comprehensive for the scope of our current analysis. However, we are open to considering this approach in future studies and appreciate your offer of the LIFE-Adult study. We would welcome further conversation with you regarding future joint projects.

Comment 2.3: "The sentences (p.17, ll.435 ff) seem to repeat: "Interestingly, we also observed a positive relationship between cortical thickness and MetS in the superior frontal, parietal and occipital lobe. Interpretation of this result is, however, less intuitive. We also noted a positive MetS-cortical thickness association in superior frontal, parietal and occipital lobes, a less intuitive finding that has been previously reported [60,61]."

Reply 2.3: Thank you for making us aware of this duplication. We have deleted the first part of the section. It now reads "We also noted a positive MetS-cortical thickness association in superior frontal, parietal and occipital lobes, a less intuitive finding that has been previously reported."

Comment 2.4: "I would highly appreciate empirical evidence for the claim in ll. 442 "In support of this hypothesis, the determined cortical thickness abnormality pattern is consistent with the atrophy pattern found in vascular mild cognitive impairment and vascular dementia" Considering the previous reports about the co-localization of obesity-associated atrophy and AD neurodegeneration (Morys et al. 2023, DOI: 10.3233/JAD-220535), that most dementias are mixed and that MetS probably increases dementia risk through both AD and vascular mechanisms, I feel such "binary" claims on VaD/AD-related atrophy patterns should be backed up empirically."

Reply 2.4: Thank you for highlighting the need for clarity in differentiating between vascular and Alzheimer's dementia. We recognize the intricate overlap in dementia pathologies. Acknowledging the prevalence of mixed dementia and the influence of MetS on both AD and vascular mechanisms, we realize our original statement might have implied a specificity to vascular dementia, which was not intended.

To address your concern, we have revised our statement to avoid an exclusive focus on vascular pathology, ensuring a more balanced representation of dementia types. Additionally, we have included Morys et al. 2023 as a reference. The section now reads: "In support of this hypothesis, the determined brain morphological abnormality pattern is consistent with the atrophy pattern found in vascular mild cognitive impairment, vascular dementia and Alzheimer's dementia."

Comment 2.5: "I wonder how specific the cell-type results are to this covariance pattern. Maybe patterns of CT (independent of MetS) show similar associations with one or more of the reported celltypes? Would it be possible to additionally show the association of the first three components of general cortical thickness variation with the cell type densities?"

Reply 2.5: Thank you for your query regarding the specificity of the cell-type results to the observed covariance pattern. To address this, we have conducted a virtual histology analysis of the first three latent variables of the main analysis PLS. The findings of this extended analysis have been detailed in the supplementary Figure S21. The imaging covariance profile of latent variable 2 was significantly associated with the density of excitatory neurons of subtype 3. The imaging covariance profile linked to latent variable 3 showed no significant association of cell type densities. Possibly, latent variable 3 represents only a noise component as it explained only 2.12% of shared variance. We hope this addition provides a clearer understanding of the specificity of our main results.

Comment 2.6: "I agree that this multivariate approach can contribute to a more holistic understanding, yet I would like to see the discussion expanded on how to move on from here. Should we target the MetS more comprehensively or would it be best to focus on obesity (being the strongest contributor and risk factor for other "downstream" conditions such as T2DM)? A holistic approach is somewhat at odds with the in-depth investigation of specific mechanisms."

Reply 2.6: We value your suggestion to elaborate on the implications of our findings. Our study indicates that obesity may have the most pronounced impact on brain morphology among MetS components, suggesting it as a key contributor to the clinical-anatomical covariance pattern observed in our analysis. This highlights obesity as a primary target for future research and preventive strategies. However, we believe that our results warrant further validation, ideally through longitudinal studies, before drawing definitive clinical conclusions.

Additionally, our study endorses a comprehensive approach to MetS, highlighting the importance of considering the syndrome as a whole to gain broader insights. We want to clarify, however, that such an approach is meant to complement, rather than replace, the study of individual cardiometabolic risk factors. The broad perspective our study adopts is facilitated by its epidemiological nature, which may not be as applicable in experimental settings that are vital for deriving mechanistic disease insights.

To reflect these points, we have expanded the discussion in our manuscript to include a more detailed consideration of these implications and future research directions.

Comment 2.7: "Please report the number of missing variables."

Reply 2.7: Thank you for your request to report the number of missing variables. We would like to direct your attention to table 1, where we have listed the number of available values for each variable in parentheses. To determine the number of missing variables, one can subtract these numbers from the total sample size.

Comment 2.8: "Was the pattern similar in pre-clinical (pre-diabetes, pre-hypertension) vs. clinical conditions?"

Reply 2.8: Thank you for your interest in the applicability of our findings across different MetS severity levels. Our analysis employs a continuous framework to encompass the entire range of vascular and cardiometabolic risks, including those only mildly affected by MetS. The linear relationship we observed between MetS severity and gray matter morphology patterns, as illustrated in Figure 2d, supports the interpretation that our findings apply to the entire spectrum of MetS severities.

Comment 2.9: "How did you deal with medication (anti-hypertensive, anti-diabetic, statins..)?"

Reply 2.9: Information on medication was considered for defining MetS for the case-control sensitivity analysis but was not included in the PLS. Detailed information can be found in table 1.

Comment 2.10: "It would be really interesting to determine the genetic variations associated with the latent component. Have you considered doing a GWAS on this, potentially in the CHARGE consortium or with UKBB as discovery and HCHS as replication sample?"

Reply 2.10: Thank you for your valuable suggestion regarding the implementation of a GWAS. We agree that incorporating a GWAS would provide significant insights, but we also recognize that it extends beyond the scope of our current analysis. However, we are actively planning a follow-up analysis. This subsequent analysis will encompass a comprehensive examination of both genetic variation and imaging findings in the context of MetS.

Comment 2.11: "Please provide more information on which data fields from UKBB were used exactly (e.g. in github repository)."

Reply 2.11: We appreciate your recommendation. The details regarding the Field IDs used in our study have been included as supplementary table S1.

Reviewer 3

Comments 3.1: "After a thorough review of the methods and results sections, I found no direct or strong evidence supporting the authors' claim that the identified latent variables were related to more severe MetS to worse cognitive performance. While a subgroup comparison was conducted, it did not adequately account for confounding factors such as educational level."

"Page 18-19 lines 431-446: the fifth paragraph in the discussion section. - As previously mentioned in the "Weaknesses" section, this study did not conduct a direct association analysis between MetS and cognitive levels without considering subgroup comparisons. Hence, I recommend the content of this paragraph warrants careful reconsideration."

Reply 3.1: We acknowledge the reviewer's constructive feedback regarding our analysis of cognitive data. We have performed a mediation analysis relating the subject-specific clinical PLS score of latent variable 1 representing MetS severity and cognitive test performances and testing for mediating effects of the imaging PLS score capturing the MetS-related brain morphological abnormalities. The imaging score was found to statistically mediate the relationship between the clinical PLS score and executive function and processing speed, memory, and reasoning test performance. These findings highlight brain structural differences as a relevant pathomechanistic correlate in the relationship of MetS and cognition. Corresponding information can now be found in figure 3, methods section 2.6.2, result section 3.3 and discussion section 4.2.

Moreover, we would like to apologize for any confusion caused by previous unclear presentation. Our study further incorporates association analyses between MetS, brain structure, and cognition using MetS components, regional brain morphological measures, and cognitive performance data in a PLS to investigate whether cognitive measures contribute to the latent variable. These analyses were separately performed on the UK Biobank and HCHS datasets, due to their distinct cognitive assessments. We adjusted for age, sex, and education in the subgroup analyses by removing their effects from the input variables. These relationships are detailed in supplementary figures S16b and S17b, with loadings close to zero for age, sex, and education, confirming effective deconfounding.

In sum, we greatly appreciate the suggestion to conduct a mediation analysis, which has substantially enhanced the strength and relevance of our analysis.

Comment 3.2: "I would suggest the authors provide a more comprehensive description of the metrics used to assess each MetS component, such as obesity (incorporating parameters like waist circumference, hip circumference, waist-hip ratio, and body mass index) and arterial hypertension (detailing metrics like systolic and diastolic blood pressure), etc."

Reply 3.2: Thank you for your suggestion regarding a more detailed description of the metrics for assessing each component of MetS. We would like to point out that the specific metrics used, including those for obesity (such as waist circumference, hip circumference, waist-hip ratio, and body mass index) and arterial hypertension (including systolic and diastolic blood pressure), are comprehensively detailed in table 1 of our manuscript. We hope this table provides the clarity and specificity you are seeking regarding the MetS assessment metrics in our study.

Comment 3.3: "I recommend the inclusion of an additional, detailed flowchart to further illustrate the procedure of virtual histology analysis. This would enhance the clarity of the methodological approach and assist readers in better comprehending the analysis method."

Reply 3.3: Thank you for your suggestion. Recognizing the challenges in visually representing many of our analysis steps, we have instead supplemented our manuscript with additional references. These references provide a clearer understanding of our virtual histology approach, particularly focusing on the processing of regional microarray expression data.

The corresponding sentence reads: "Further details on the processing steps covered by ABAnnotate can be found elsewhere (<https://osf.io/gcxun>) [42]"

Comment 3.4: "Why were both brain hemispheres used instead of solely utilizing the left hemisphere as the atlas, especially considering that the Allen Human Brain Atlas (AHBA) only includes gene data for the right hemisphere for two subjects?"

Reply 3.4: Thank you for your query regarding our decision to use both brain hemispheres instead of solely the left hemisphere, especially considering the Allen Human Brain Atlas (AHBA) predominantly featuring gene data from the left hemisphere. Given the AHBA's limited spatial coverage of expression data in the right hemisphere, our approach involved mirroring the existing tissue samples across the left-right hemisphere boundary using the abagen toolbox,¹ a practice supported by findings that suggest minimal lateralization of microarray expression.^{2,3} Further details are provided in previous work employing ABAnnotate.⁴ These studies are now referenced in our methods section.

Comment 3.5: "The second latent variable was not further discussed. If this result is deemed significant, it warrants a more detailed discussion. "

Reply 3.5: Thank you for the suggestion. We have added a paragraph to the discussion that discusses the second latent variable in greater detail. It reads: "The second latent variable accounted for 22.33% of shared variance and linked higher insulin resistance and lower dyslipidemia to lower thickness and volume in lateral frontal, posterior temporal, parietal and occipital regions. The distinct covariance profile of this latent variable, compared to the first, likely indicates a separate pathomechanistic connection between MetS components and brain morphology. Given that HbA1c and blood glucose were the most significant

contributors to this variable, insulin resistance might drive the observed clinical/anatomical relationship.”

Comment 3.6: “I suggest appending positive MetS effects after “..., insular, cingulate and temporal cortices;” for two reasons: a). The “positive MetS effects” might represent crucial findings that should not be omitted. b). Including both negative and positive effects ensures that subsequent references to “this pattern” are more precise.”

Reply 3.6: We concur with the notion that the positive MetS effects should be highlighted as well. We modified the first discussion paragraph now mentioning them.

Comment 3.7: “I would appreciate further clarification on this sentence and the use of the term “uniform” in this context. Does this suggest that despite the heterogeneity in the physiological and pathological characteristics of the various MetS components (e.g., obesity, hypertension), their impacts on cortical thickness manifest similarly? How is it that these diverse components lead to “uniform” effects on cortical thickness? Does this observation align with or deviate from previous findings in the literature?”

Reply 3.7: Thank you for highlighting the ambiguity in our previous explanation. We agree that the complexity of the relationship between MetS components and brain morphology requires clearer articulation. To address this, we have revised the relevant sentence for better clarity. It now reads: „This finding indicates a relatively uniform connection between MetS and brain morphology, implying that the associative effects of various MetS components on brain structure are comparatively similar, despite the distinct pathomechanisms each component entails.“

Comment 3.8: “Figure 1 does not have the labels “c)” and “d)”. ”

Reply 3.8: Thank you. We have modified figure 1 and made sure that the caption correctly references its content.

Comment 3.10: “Incorrect figure/table citation:

- *Page 18 line 418: “(figure 2b and 1c)” à (figure 2b and 2c).*
- *Page 18 line 419: “(supplementary figures S8 and S12-13)” à (supplementary figures S11 and S1516).*
- *In the supplementary material, “Text S5 - Case-control analysis” section contains several figure or table citation errors. Please take a moment to review and correct them.”*

Reply 3.10: Thank you for bringing this to our attention. We have corrected the figure and table citation errors.

Comment 3.11: “Page 8 line 184: The more commonly used term is “insulin resistance” rather than “insuline resistance.”

Reply 3.11: We now use “insulin resistance” throughout the manuscript.

Comment 3.12: “Nevertheless, variations in gene sets may introduce a degree of heterogeneity in the results (Seidlitz, et al., 2020; Martins et al., 2021). Consequently,

further validation or exploratory analyses utilizing different gene sets can yield more compelling results and conclusions."

Reply 3.12: Thank you for your insightful comment regarding the potential heterogeneity introduced by variations in gene sets. We agree that exploring different gene sets could indeed enhance the robustness and generalizability of our findings. However, we think conducting a comprehensive methodological analysis of the available cell-type specific gene sets is a substantial effort and warrants its own investigation to thoroughly implement it and assess its implications. We also like to highlight that we are adhering to previous practices in our analysis setup.^{4,5}

References

- (1) Markello RD, Arnatkeviciute A, Poline JB, Fulcher BD, Fornito A, Misic B. Standardizing workflows in imaging transcriptomics with the abagen toolbox. Jbabdi S, Makin TR, Jbabdi S, Burt J, Hawrylycz MJ, eds. eLife. 2021;10:e72129. doi:10.7554/eLife.72129
- (2) Hawrylycz MJ, Lein ES, Guillozet-Bongaarts AL, et al. An anatomically comprehensive atlas of the adult human brain transcriptome. Nature. 2012;489(7416):391-399. doi:10.1038/nature11405
- (3) Hawrylycz M, Miller JA, Menon V, et al. Canonical genetic signatures of the adult human brain. Nat Neurosci. 2015;18(12):1832-1844. doi:10.1038/nn.4171
- (4) Lotter LD, Saberi A, Hansen JY, et al. Human cortex development is shaped by molecular and cellular brain systems. Published online May 5, 2023:2023.05.05.539537. doi:10.1101/2023.05.05.539537
- (5) Lotter LD, Kohl SH, Gerloff C, et al. Revealing the neurobiology underlying interpersonal neural synchronization with multimodal data fusion. Neuroscience & Biobehavioral Reviews. 2023;146:105042. doi:10.1016/j.neubiorev.2023.105042